JNET Communication

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1.1 Additional Information

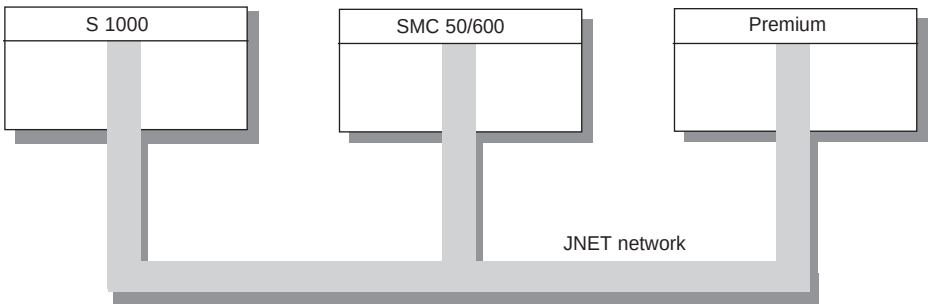
Other documentation may be necessary when installing the JNET module.
Manuals :

- **TSX DM57 F/N,** part D, presents general information relating to the hardware aspects of network installation.
- **TLX DSCOM PL7** presents all general information relating to the software installation for different networks.
- **TEM60105F** presents all information relating to the installation of JNET modules on APRIL® 5000 and APRIL® 7000 PLCs.
- **TEM15060F** presents all information relating to the installation of JNET modules on APRIL® 2000 and APRIL® 3000 PLCs.
- **F-5JNET** presents all information relating to the installation of JNET modules on SMC 50/600 PLCs.

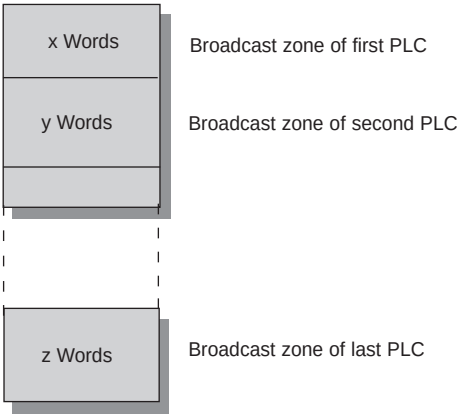
1.2 Presentation

1.2-1 Jnet Protocol

The JNET function is an exchange of data between Series 1000, SMC50/600 and Premium PLCs. Bus topology used is deterministic token passing.



The total number of words exchanged constitutes a table.
This table resides in each PLC and is divided into the same number of zones as the number of PLCs on the JNET network. The zone allocated to each PLC varies in size (defined during configuration).



The Jnet protocol ensures that the table is updated in each PLC independently of the PLC program.

Compatibility : April 2000 - April 3000 - April 5000 - April 7000 - SMC50/600 - Premium

Number of PLCs : 32 maximum (16 for an SMC type network)

Fixed transmission speed : 19200 bauds

Fixed transmission format : 8 bits, no parity, 1 stop bit

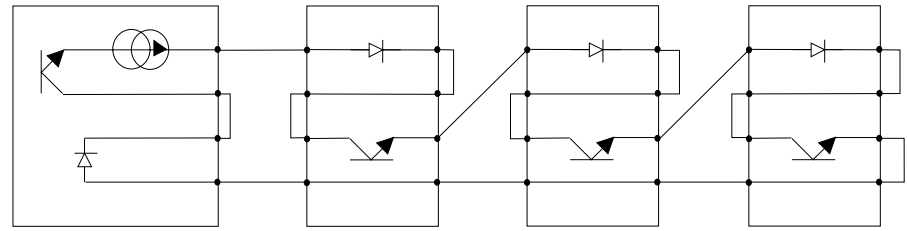
Data transmitted : 128 words maximum shared between all PLCs (64 words max. for an SMC type network)

Transmission medium : current loop or 2-wire RS485 link

1.2-2 Physical Interfaces

20 mA current loop link

- Transmission is via a single shielded twisted pair.
- The idle condition of the line is 20 mA.
- Number of stations ≤ 16.
- All transmitters and receivers on a JNET network are connected in series on the same loop. The schematic diagram is as follows (example with 4 stations) :



Station 1 (supplies the line) Station 2 Station 3 Station 4

RS485 link

- Differential transmission mode (2 wires)
- Direct link using 1 shielded twisted pair, characteristic impedance 120 Ω
- Line terminators
- Number of stations ≤ 32
- Network length ≈ 1.3 km
- Length of tap links ≤ 15 meters

1.2-3 Compatibility

Jnet on Premium PLCs can be configured for both types of network :

- 16 stations / 64 words max. in the shared table
- 32 stations / 128 words max. in the shared table

The module can consequently be connected to a network which includes SMC or S1000 PLCs.
It may not be compatible with PB PLCs.

Compatibility with existing models can be summarized as follows :

Type of device	Type of network		Transmission medium	
	16 stations / 64 words	32 stations / 128 words	Current loop	RS485
SMC	X		X	
S1000	X	X	X	X
Premium	X	X	X	X

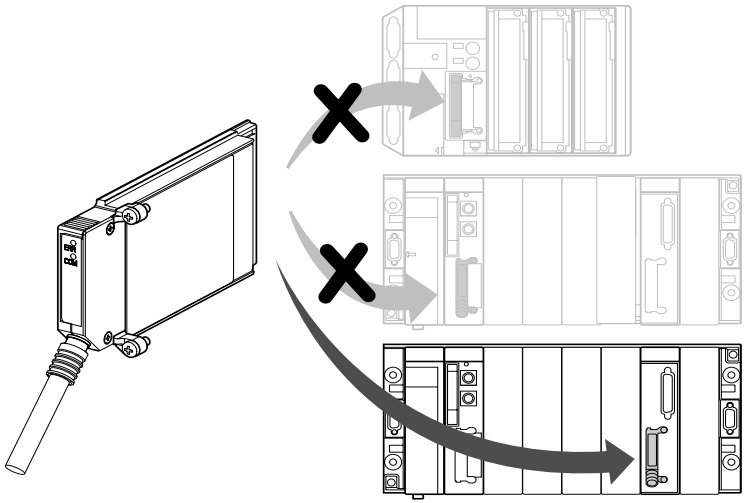
JNET modules can be configured using the following PL7 versions : **V3.0 and ≥**

1.3 Hardware Installation

TSX Premium PLCs are connected to the networks, bus and data communication link using PCMCIA communication cards.
The connected card consists of a metal case with dimensions conforming to the type III extended PCMCIA format.

The Jnet function is in the form of two cards in type III extended PCMCIA format :

- TSX JNP 112 which supports a physical layer consisting of a 20 mA current loop link
- TSX JNP 114 which supports a physical layer consisting of an RS485 link



The PCMCIA cards only use the Jnet protocol and **can only be installed in the relevant slot in the TSX SCY 21601 module of TSX Premium family PLCs**.

Comment :

Under no circumstances handle PCMCIA cards when powered up.

The cards are installed and operated using PL7 Junior programming and operating software for TSX Premium PLCs.

1.4 Limits

Number of JNET modules per PLC : Jnet is an asynchronous communication channel which the system can assign as a Jbus or Unitelway communication channel. It is therefore counted as an "expert channel".
A maximum of three JNET channels can be installed in one PLC.

1.5 Operating Standards

The **TSX SCY 21601** module and **PCMCIA** communication cards conform to the following international standards :

- US standards : UL508, IEC 1131-2
- Canadian standards : CSA C22.2 / 142
- Conform to regulation : FCC-B
- CE marking
- Type III E PCMCIA mechanical standard
- PCMCIA 2.01

2.1 Mounting PCMCIA Cards

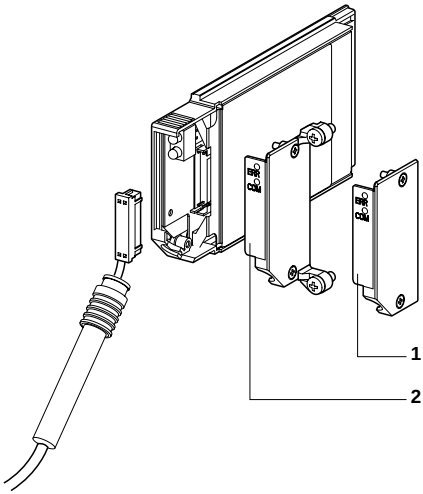
Type III E (extended) PCMCIA communication cards are integrated in a metal case with the following dimensions :

- length 85.6 mm
- width 51 mm
- height 10 mm

The front panel of the card displays communication operations and the physical network connection.

The mechanical configuration of the card can be adapted to the type of installation required. The PCMCIA card is mounted either with a removable cover fitted with two lugs, or with a plain removable cover.

The plain removable cover (**1**) is for connecting PC-compatible devices.
The removable cover with lugs (**2**) has fixing screws for physically linking it to the host module (PCMCIA card inserted in the TSX Premium processor for example).



Note : Only the removable cover with lugs (2) is supplied with a PCMCIA JNET card which can only be mounted on an SCY module for the TSX Premium PLC.

The card is connected to the network via the connection cable on the front panel. A locating system avoids incorrect mounting. The product reference label informs the user of the type of physical layer supported by the card.

Note
The use of covers with lugs on PCMCIA cards avoids any accidental removal of the card while powered up and ensures that the card operates correctly.

2.2 Connecting the PCMCIA Card Host Channel

2.2-1 Precautions When Connecting the PCMCIA Card

The PCMCIA card **must be handled when powered down**. Operation of the unit is not guaranteed when inserting or removing the card. There is no warm restart procedure for the PCMCIA card when used with the TSX SCY 21601 host device.

If, owing to the operating environment, the application cannot be stopped by powering down the PLC processor, we recommend that you remove the TSX SCY 21601 module with the PCMCIA card.

The PCMCIA card must be equipped with the appropriate cover for the PLC version used and be screwed into the TSX SCY 21601 host module before powering up the unit, as described in the installation manual for TSX 57 PLCs (ref. TSX DM 57F/N, part D).

2.2-2 PCMCIA Card References

The PCMCIA card references are as follows :

- **TSX JNP 112** : JNET protocol card, 20 mA current loop
- **TSX JNP 114** : JNET protocol card, RS 485

2.2-3 Connecting PCMCIA Cards

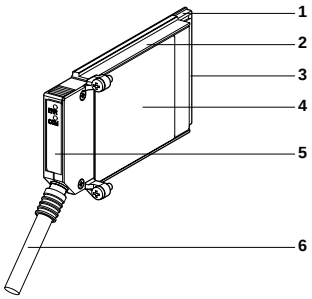
List of required connection devices :

PCMCIA cards	Cables	Junction boxes
TSX JNP 112 (CL)	TSX SCP CX 2030 **II ≥03**	1 BJ 110/220 or 1 BJ 24 (according to wiring)
TSX JNP 114 (RS 485)	TSX SCP CM 4030 **II ≥01**	TSX SCA 50 or TBX 0010

2.2-4 Mounting Cards and Cables for the TSX PREMIUM

PCMCIA cards consist of the following elements :

1 Ready assembled card
2 Zinc body
3 PCMCIA connector
4 Upper cover
5 Removable cover
6 Connection cable with ferrule



The removable cover (**5**) displays the operation of the card in its environment. The description of the two indicator lamps is printed on the front panel of the removable cover.

The product reference label indicates the type of PCMCIA card. It is attached to the upper cover (**4**).

The metal ferrule (**6**) attached to the cable at the PCMCIA card end prevents the cable from being constricted by the removable cover. The ferrule prevents any risk of curvature in the cable which might reduce the quality of the link.

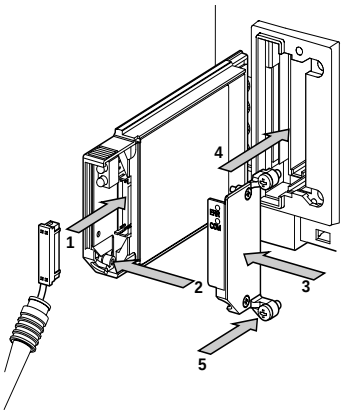
The PCMCIA card is mounted by assembling the connection accessory (the cable type varies depending on the transmission medium selected), then screwing the fixing lugs of the removable cover to the junction box. The cover enables the PCMCIA card to be fixed in the TSX SCY 21601 module.

There is a 20-pin connector at the PCMCIA card end.

To connect the transmission medium to the card, first remove the cover screwed onto the junction box, then proceed to mount it as follows :

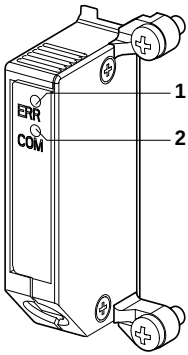
1 Connect the cable.
2 Place the cover with lugs on the junction box.
3 Screw on the cover.
4 Insert the card in the slot provided for this purpose in the host device.
5 Screw the card in to avoid any handling of the card while powered up and to ensure that it operates correctly.

Host
TSX SCY 21601



2.2-5 Displaying the Operation of PCMCIA Cards

Two diagnostic indicator lamps are located on the front panel of the card. They inform the user of the operation of exchanges between the device supporting the PCMCIA card and the connected device.



The Error indicator "ERR" (**1**) (normally off) displays faults.

The "ERR" indicator is red.

The Communication indicator "COM" (**2**) displays line activity (transmission/reception).

The "COM" indicator is yellow.

2.2-6 Visual Diagnostics of PCMCIA Cards

The status of the PCMCIA card indicator lamps indicates the communication operating mode and the card diagnostics.

Lamp status

ERR	COM	Meaning	Corrective action
○	○	• Device off • No dialog May be due to an overlap problem (diagnostics given by module)	Check PLC power supply and connection Correct JNET network configuration and restart via switch off / switch on or by reloading the program
○	◌	Normal operation	
●	(1)	Serious fault	Change card
◌	○	Operational fault	Check configuration and connection to PLC bus (possible problem with CPU ↔ JNET module dialog)
◌	◌	Operational fault	Check configuration

● Indicator lamp on ○ Indicator lamp off ◌ Indicator lamp flashing

(1) : Indicator lamp status not applicable

2.3 Connecting the TSX JNP 112 Card

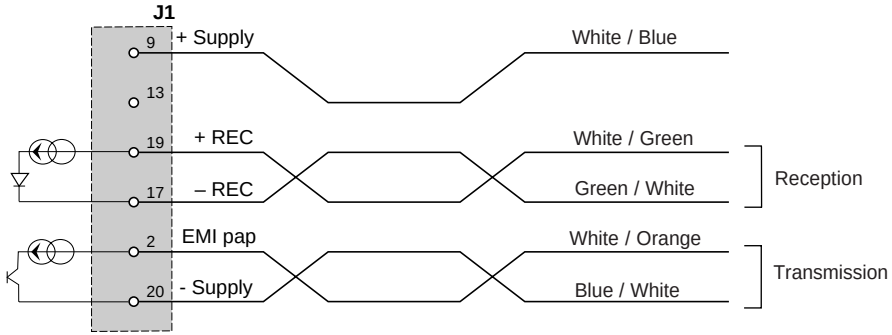
The TSX JNP 112 PCMCIA card is used to connect a TSX 57 PLC to a JNET network using a 20mA current loop link.

In all cases an external power supply to the TSX JNP 112 card must provide the current required for the current loop power supply.

The TSX SCP CX 2030 cable can be used for this type of connection (length 3m).

Description of the TSX SCP CX 2030 cable

The 20-pin PCMCIA mini-connector supports the following signals :



Notes

The TSX JNP 112 card connection requires the installation of a screw terminal block.

Preliminary comments

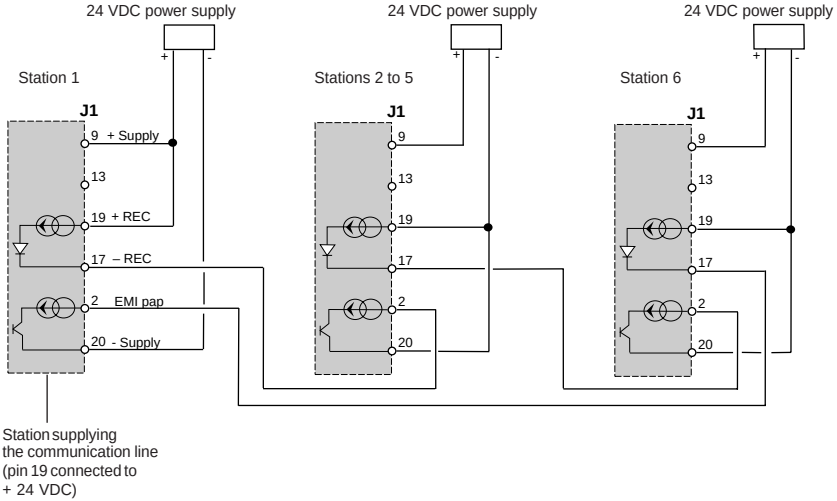
- In a JNET network, transmitters and receivers are connected in series on the same loop.
- The idle condition of the line is 20 mA.
- There must be a 24V power supply for the current loop (supplied from a single station).
- There must be a 24V power supply to each PCMCIA card and **these power supplies must be floating with no common point** (for example : ground).

2.3-1 Connecting to the JNET Network Without a Junction Box

This section gives wiring examples of configurations with a small number of stations where network availability is not critical.

Characteristic : max. 6 stations

• Example with 3 PREMIUM stations

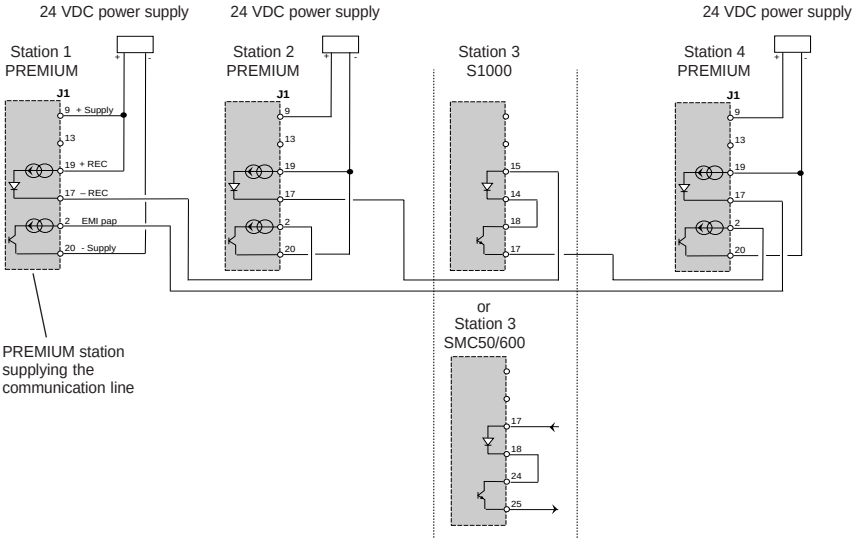


Terminal	Designation	Wire colors
9	+ Supply	White/blue
19	+ REC	White/green
17	- REC	Green/White
2	EMI pap	White/Orange
20	- Supply	Blue/White

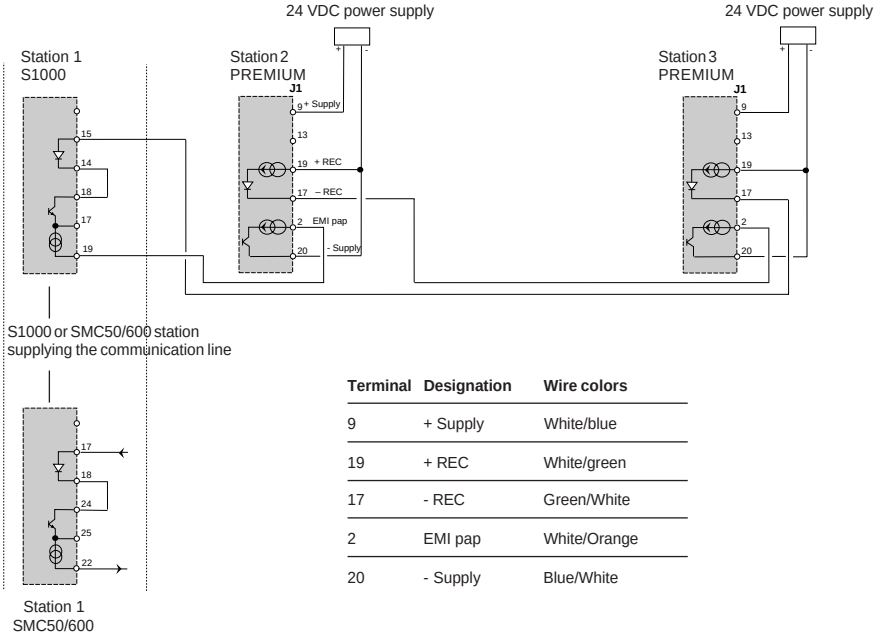
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• Examples with PREMIUM and S1000 or SMC50/600 stations

- PREMIUM station polarizing the line



- S1000 or SMC50/600 station polarizing the line



Terminal	Designation	Wire colors
9	+ Supply	White/blue
19	+ REC	White/green
17	- REC	Green/White
2	EMI pap	White/Orange
20	- Supply	Blue/White

2.3-2 Connecting to the JNET Network with a Junction Box

This section gives some wiring examples for increasing the number of stations and disconnecting certain PLC clusters.
The junction box must be used with more than 6 stations.

Junction box references: 1 BJ 110/220
1 BJ 24

Characteristics : - 16 stations which can be connected to each other.
- Maximum of 6 stations per B cluster.
- The A cluster is connected to the master end of the junction box.
- The B clusters are connected to the slave ends of the junction box. These clusters can be powered down or disconnected without disturbing network operation.
- The master end of the junction box must always be wired or strapped.

Diagram 1

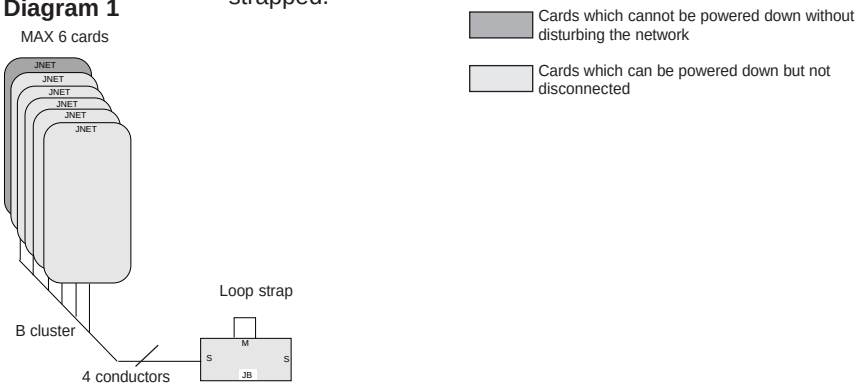


Diagram 2

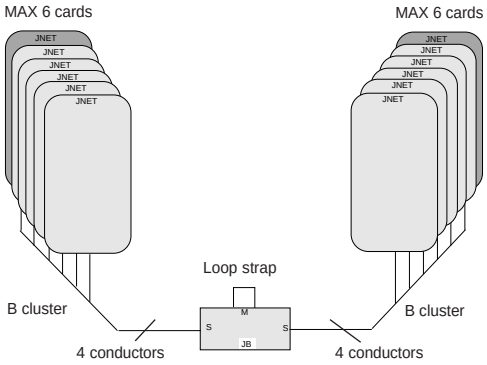


Diagram 3

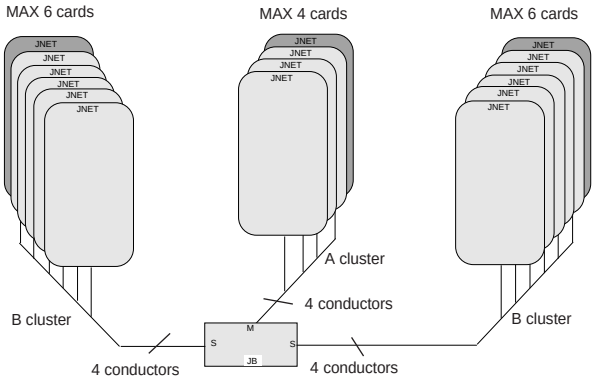
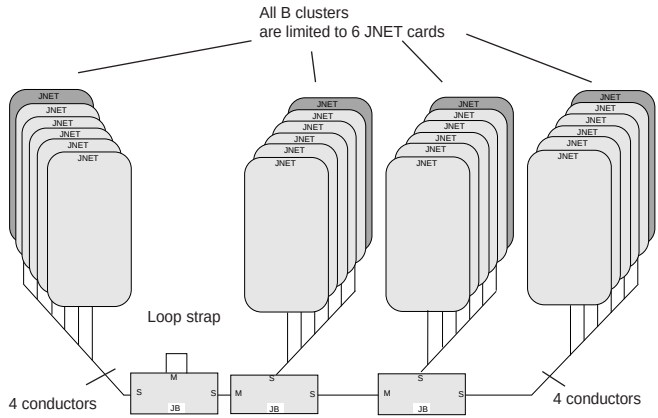


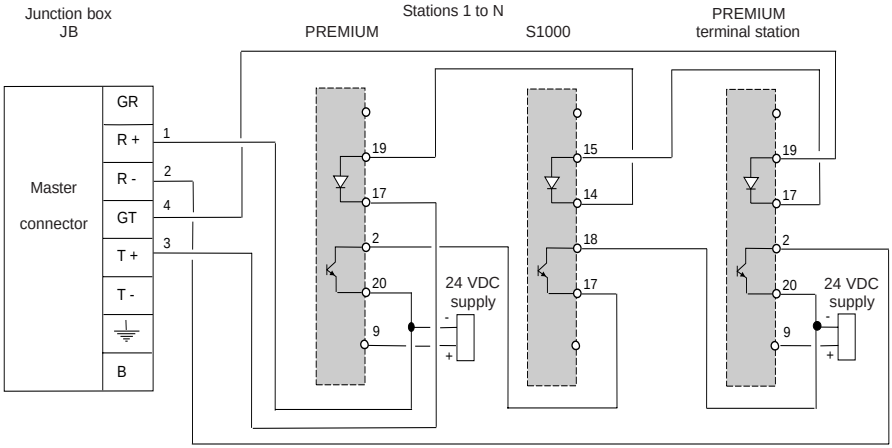
Diagram 4



Comments :

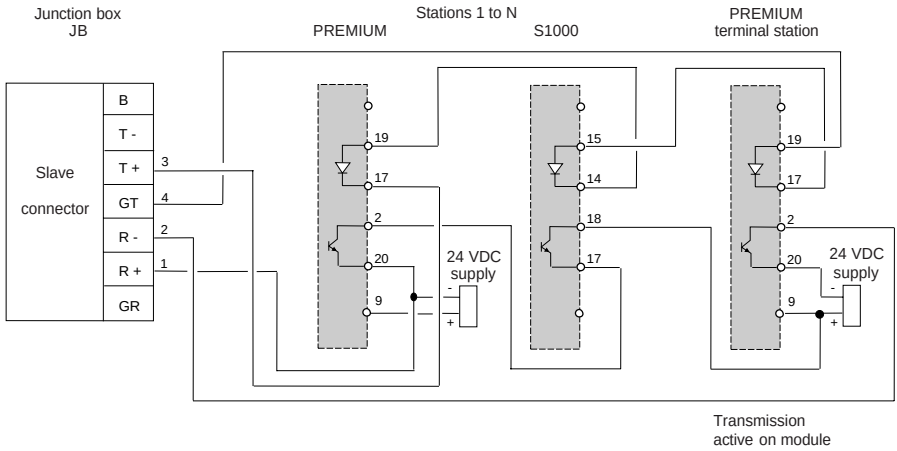
- The JNET cards represented can be for Premium, S1000 or SMC50/600 PLCs.
- No more than two PREMIUM stations may be in one cluster.

• Connecting an A cluster

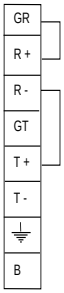


Terminal	Designation	Wire colors
9	+ Supply	White/blue
19	+ REC	White/green
17	- REC	Green/White
2	EMI pap	White/Orange
20	- Supply	Blue/White

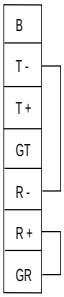
• Connecting a B cluster



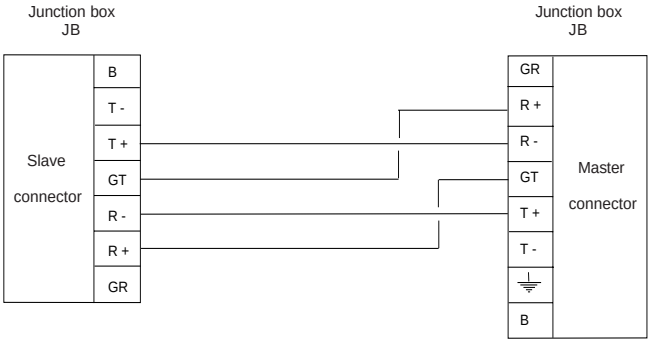
• Loop strap at the master end of junction box "JB"



• Loop strap for a slave connector (optional on a free connector)

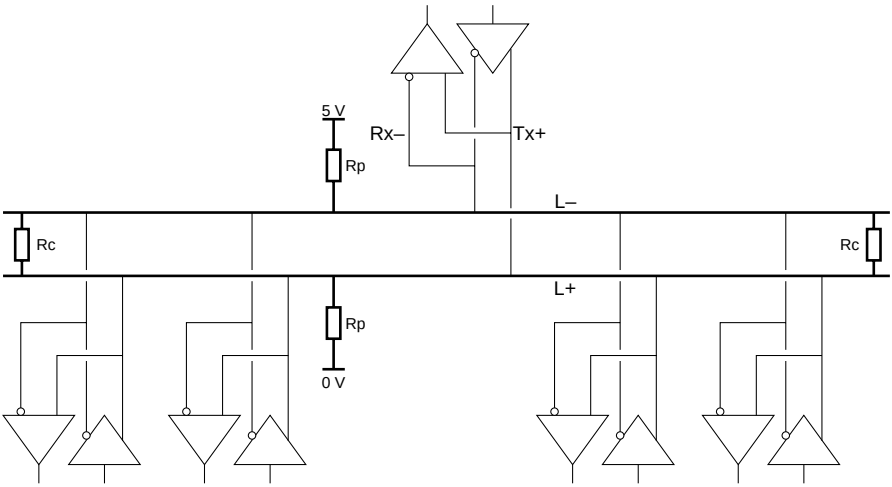


• Connecting wiring between 2 junction boxes "JB"



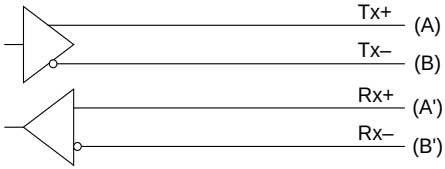
2.4 Connecting the TSX JNP 114 Card

2.4-1 Summary of Single RS 485 Line Termination



The diagram above shows the general architecture of an RS 485 network.

Transmitters are symbolized by :



Receivers are symbolized by :

The network is made up of a single shielded twisted pair. The various network stations are connected simply by connecting :

- all outputs marked + (Tx+, Rx+) to the + wire of the network marked (L+)
- all outputs marked - (Tx-, Rx-) to the - wire of the network marked (L-)

The network impedance is adapted using two impedance matching resistors (Rc) located on the two end stations of the network.

The network is polarized by connecting the L+ wire to the 0 V and the L- wire to the 5 V via the two polarization resistors (R = 470Ω). The polarization causes a current to flow continuously through the network. This adaptor can be located anywhere on the network.

It must be unique for the entire network irrespective of the network length. For maintenance purposes, it is advisable to mark the cable of the station performing the polarization function for easy identification.

2.4-2 Characteristics

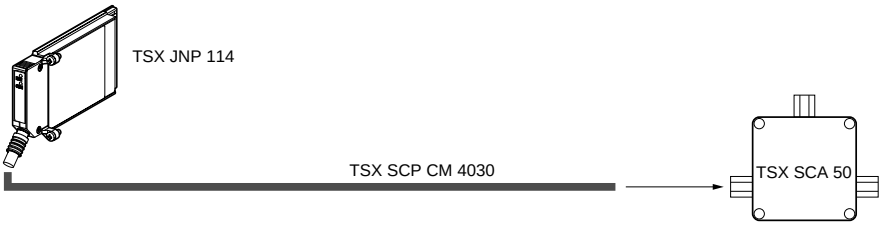
- 1 shielded twisted pair
- Maximum 32 stations
- Maximum length : 1300 m approx.
- Bus topology
- Tap link <= 15 m
- Half duplex on 2 wires
- Line terminator fitted to end stations
- Unique line polarization Rp = 470 Ω

2.4-3 Connecting to the JNET Network

The TSX JNP 114 PCMCIA card is connected to the JNET network using the TSX SCP CM 4030 serial connection cable. This cable is connected to the TSX SCA 50 T-junction box or the TBX 0010 junction box.

Type of connection

The cable of the PCMCIA card has bare wires at the end for connection to the terminal block located inside the junction box.

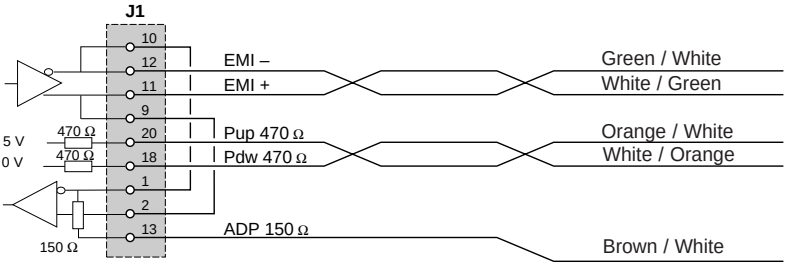


Comments :

- The length of the user cable (3 m) enables a device to be connected to a TSX SCA 50 or TBX 0010 junction box located within a 3 meter radius of the card. This length ensures that it can be connected inside a standard enclosure.
- The cable shielding is grounded via the PCMCIA card.

Description of TSX SCP CM 4030 cable

The 20-pin PCMCIA mini-connector
supports the following signals :



Notes :

• For an intermediate station, only EMI+ and EMI- should be connected to the network.

• For the two end stations, the impedance matching resistor ADP must be connected between EMI+ and EMI- (the card makes the internal connection to EMI-).The wires from terminals 13 and 2 must be connected.

• On one station (only) the line must be polarized using the 470 Ω Pull-up and Pull-down resistors on the PCMCIA card. Connect Pup (terminal 20) to EMI+ (terminal 11) and Pdw (terminal 18) to EMI- (terminal 12)

Important :

When connecting to S1000 PLCs, EMI+ must be connected to L- (Rx-/Tx-) and EMI- must be connected to L+ (Rx+/Tx+).

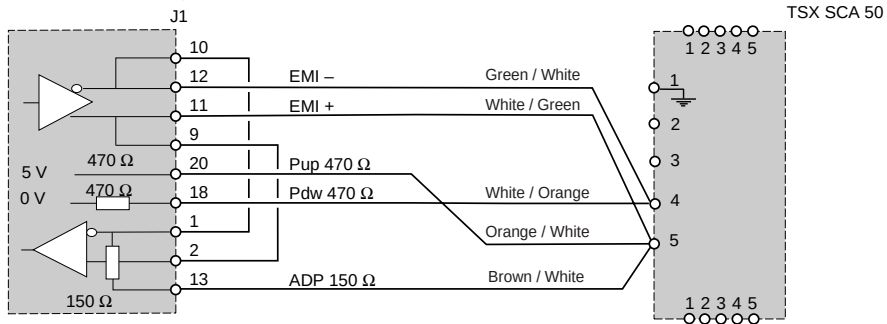
ENGLISH

• Examples of wiring using the TSX SCA 50 T-junction box

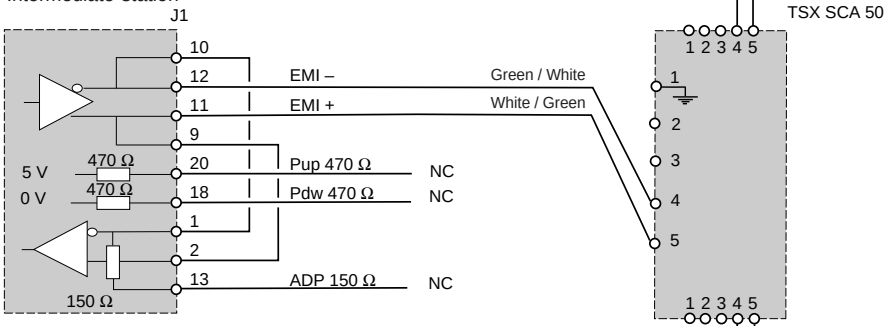
Do not use the impedance matching resistor on the TSX SCA 50 T-junction box. The jumper must be in the OFF position.

- PREMIUM PLCs network only

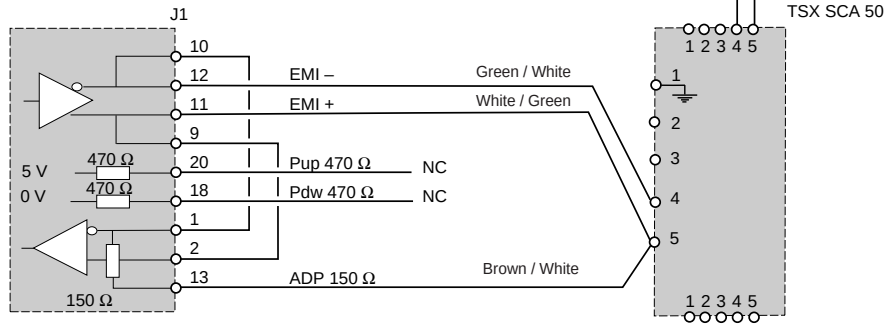
End station and line polarization



Intermediate station



End station



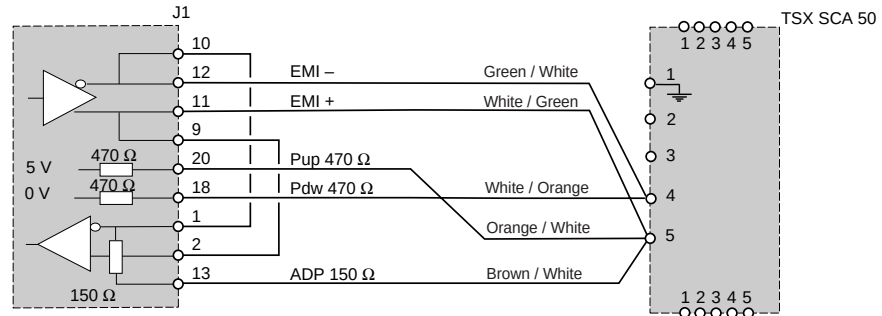
NC : Not connected

- PREMIUM and S1000 PLCs network

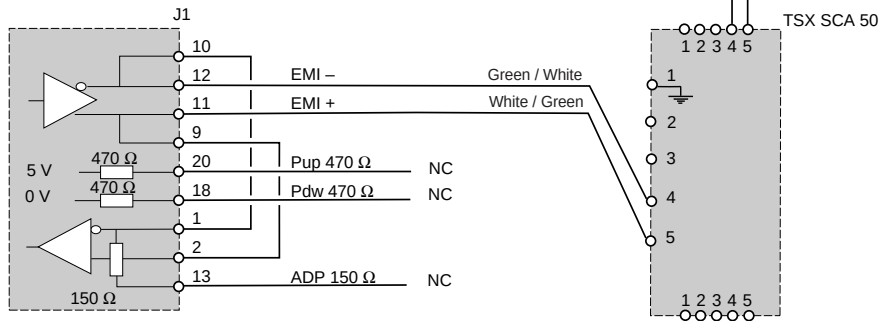
Comment :

EMI+ must be connected to L- and EMI- must be connected to L+

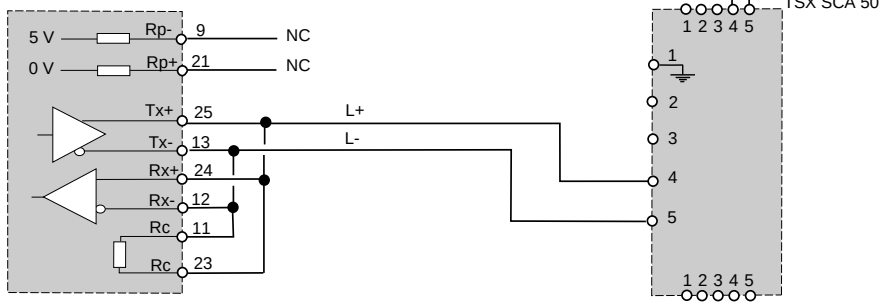
End station and line polarization (PREMIUM)



Intermediate station (PREMIUM)



End station (S1000)



NC : Not connected

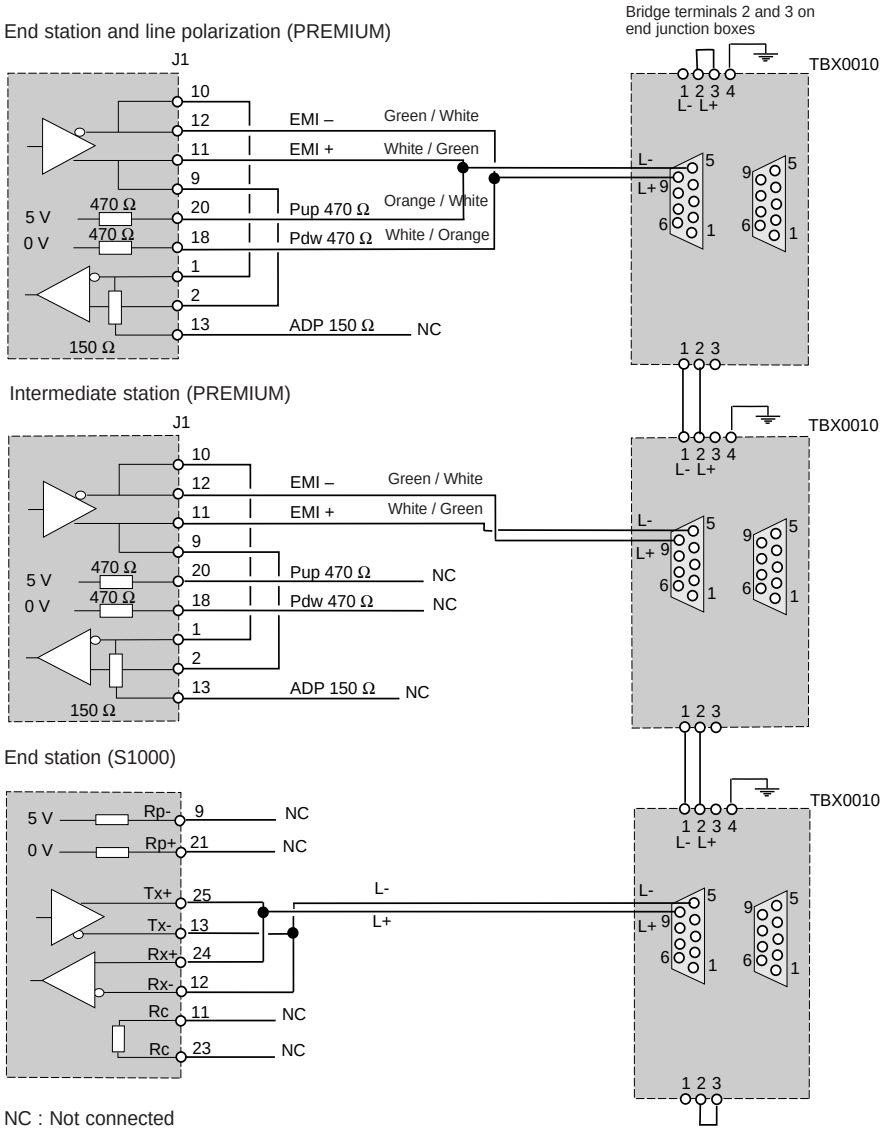
ENGLISH

• Examples of wiring using the TBX0010 junction box

The TBX0010 junction box has an impedance matching resistor for use on the end junction boxes via a strap.

Comment :

EMI+ must be connected to L- and EMI- must be connected to L+.



NC : Not connected

2.5 Summary of Connection Cables

2.5-1 TSX JNP 112 Card

Type of cable	Reference	Description
Current loop cable	TSX SCP CX 2030 **II≥03**	20mA CL cable L = 3m.

2.5-2 TSX JNP 114 Card

Type of cable	Reference	Description
Modbus/JNET cable	TSX SCP CM 4030 **II≥01**	RS 485 2-wire cable L = 3 m
T-junction box	TSX SCA 50	T-junction with screw connection to bus for RS 485 serial link
Junction box	TBX 0010	Junction box with SUB-D connectors for RS 485 serial link

2.6 Recommendations When Connecting PCMCIA Cards

Important

PCMCIA cards must be connected to and disconnected from the host device (TSX SCY 21601) with the device **powered down**.

The ferrule placed directly in contact with the PCMCIA card junction box enables electrical interference carried by the connection cable braid to be eliminated.

2.7 Power Consumption of PCMCIA Cards

2.7-1 Power Consumption of TSX JNP 112 Card

Voltage	Typical current	Maximum current
5 volts	120 mA	300 mA

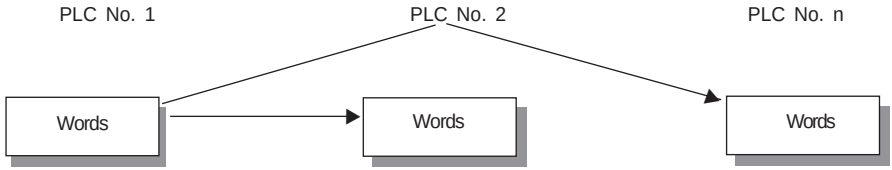
2.7-2 Power Consumption of TSX JNP 114 Card

Voltage	Typical current	Maximum current
5 volts	150 mA	300 mA

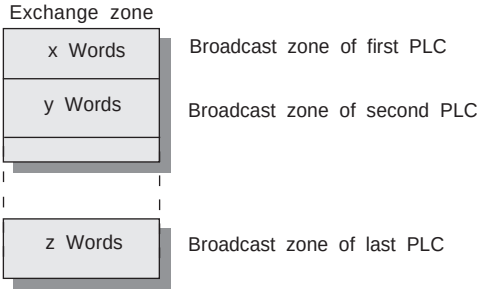
3.1 Presentation of the JNET Function

3.1-1 Principle of JNET Exchanges

JNET exchanges are based on the principle that each PLC broadcasts a word memory zone (broadcast or local zone) to the other PLCs in the network.

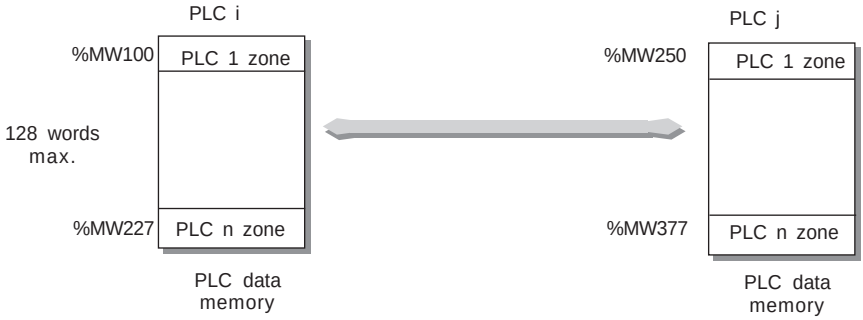


The total number of words exchanged constitutes a table : **Exchange zone.**



This zone must be defined in each PLC and have the same length in each PLC.

Example :



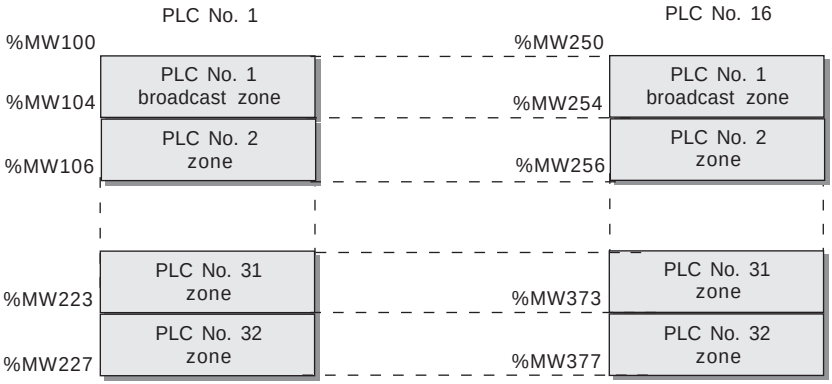
3.1-2 Setting Exchange Zone Parameters

The exchange zone must be included in the memory zone for type %MW data (16-bit words).
The characteristics of this table are given in the JNET configuration screen.

The exchange table is divided into local broadcast zones.

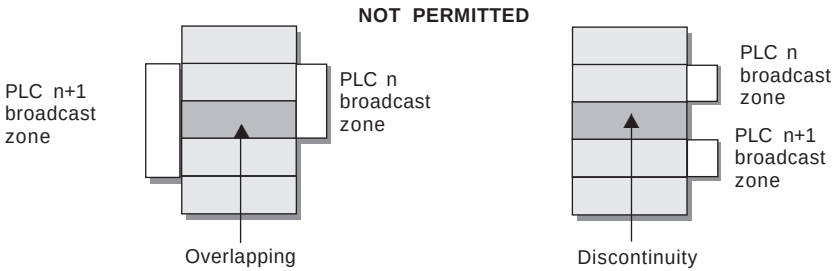
- The size of the exchange table is limited to 128 words maximum.
- The size of the local zone is limited to 64 words maximum and must contain 1 word as a minimum.
- The size of the exchange table reserved in a station is equal to the sum of the sizes of the zones produced by each station.
- The broadcast zones must be contiguous and in ascending order by station number.
- Each broadcast zone is assigned to 1 PLC.
- The broadcast zone for PLC x must have the same length in all PLCs.

Example



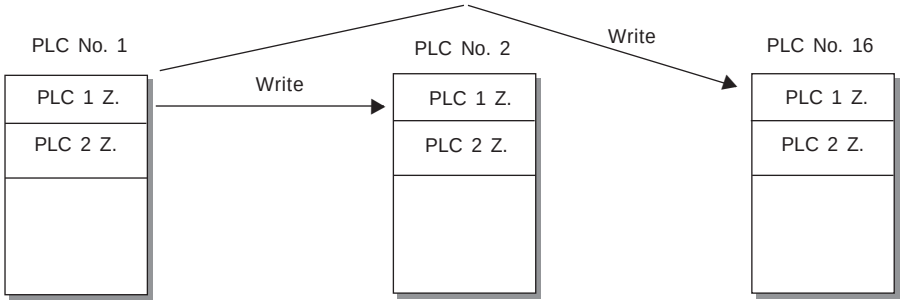
Caution

The broadcast zones must be contiguous (no overlapping or discontinuity).



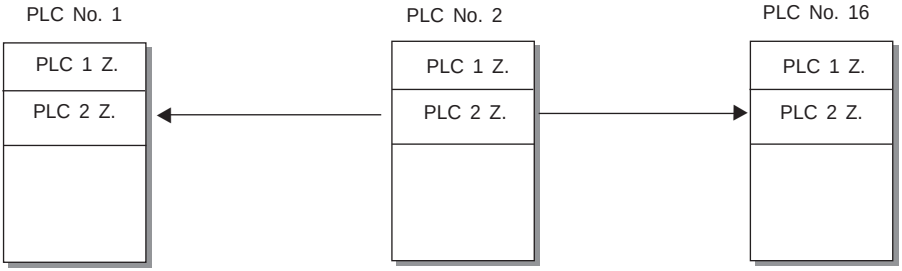
3.1-3 Broadcasting Data

PLC no. 1 broadcasts data from its broadcast zone to all PLCs.



- PLC n Z. : broadcast zone assigned to PLC n

PLC No. 2 then broadcasts data from its broadcast zone to all PLCs.



PLC No. 3 then executes this procedure and so on through to the last PLC. It is then the turn of PLC No. 1 again.

For a 16 station configuration the maximum exchange of 64 words between all 16 PLCs takes approximately 110 ms on the network.
This applies if all stations send data simultaneously.

- Each JNET card only sends its broadcast zone if the content has changed since the last transmission : the card sends a "data frame".
- If the content has not changed, the JNET card only transmits a right-to-send or a TOKEN for the next station.
- Additional information on exchange times is available in the appendix.

3.1-4 Updating Data in the CPU

The JNET function ensures continuous data exchange between network stations. This data circulates on the network in a way that is totally transparent to the user.
Data from other stations is stored in the module as it is received.
The PLC memory is updated as frequently as possible on the initiative of the CPU, via message exchanges at the end of the program scan (optimally on each program scan).
Data exchange between the CPU and the module is automatic and not conditioned by any change in value: this ensures data consistency between the CPU and the JNET module.

On CPU start-up the word zone assigned to the JNET network is either :

- not initialized
- at zero

Therefore, as long as there has not been a JNET network cycle to update all data in the modules and a program scan to transfer this data to the CPU, the JNET memory zone in the CPU is not significant : **as a precaution, the use of JNET data during the first PLC cycles must be suspended.**

3.2 Language Interface

The various language objects associated with communication via JNET are shown in the table below. They complement the language objects common to all communication modules.

Object	Function	Meaning
%Im.1.ERR I	Channel fault	= 1 if line failure (one of the faults sent to %MWm.1.2). This bit activates %Im.MOD.ERR
%IWm.1.0 I	Special channel status (linked to JNET protocol)	Byte 0 = 0 : Normal operation Byte 0 = 1 : At least one station is in stop mode Byte 0 = 2 : At least one station is absent Byte 0 = 3 : The station is synchronizing itself with the network Byte 0 = 4 : Overlapping of broadcast zones (signals overlapping or discontinuity of zones) Byte 0 = 5 : Echo fault (line break)
%MWm.1.0 E	Exchange in progress	x0 = 1 : Read current status x1 = 1 : Send control parameters to the communication channel x2 = 1 : Send adjustment parameters to the communication channel (there are none for the JNET function)
%MWm.1.1 E	Report	x1 = 0 : Control parameters received and accepted by the communication channel x2 = 0 : Adjustment parameters received and accepted by the communication channel (there are none for the JNET function)
%MWm.1.2 E	Standard channel status	x0 to x3 : not used x4 = 1 : Module faulty, absent or self-tests in progress or promptitude fault x5 = 1 : Hardware or software configuration fault x6 = 1 : Communication fault x7 = 1 : Application program fault (error in configuration, etc)
%MWm.1.3 E	Fault counter	CRC
%MWm.1.4 E	Special status	Status of stations 1 to 16 (1 bit per station) : bit = 0 : station OK (or not configured) bit = 1 : station /OK : see details in word MWm.1.6
%MWm.1.5 E	Special status	Status of stations 17 to 32 (1 bit per station) : bit = 0 : station OK (or not configured) bit = 1 : station /OK : see details in word MWm.1.7

Notes : I = **I**mplicit exchanges E = **E**xplicit exchanges m = **m**odule position

Object	Function	Meaning
%MWm.1.6 E	Special status	Complement to status of stations 1 to 16 (1 bit per significant station if the corresponding bit in %MWm.1.4 is at 1) : • bit = 0 : station in stop mode • bit = 1 : station absent
%MWm.1.7 E	Special status	Complement to status of stations 17 to 32 (1 bit per significant station if the corresponding bit in %MWm.1.5 is at 1) : • bit = 0 : station in stop mode • bit = 1 : station absent
%MWm.1.8 E to %MWm.1.14	Reserved	
%MWm.1.15E	Command	x0 = 1 : Reset counter

Configuration data

%KWm.1.0 I	Channel type	JNET function = h'0D'
%KWm.1.1 I	Network type / Operating mode	Byte 0 = 0 : 16 stations - 64 words Byte 0 = 1 : 32 stations - 128 words Byte 1 = 0 : Reset zone for stations in stop mode or absent Byte 1 = 1 : Freeze zone for stations in stop mode or absent
%KWm.1.2 I	Number of stations	≤ 16 or 32 according to network type
%KWm.1.3 I	Station number	less than or equal to number of stations
%KWm.1.4 I	Exchange zone start address	
%KWm.1.5 I	Total length of exchange zone	≤ 64 or 128 according to network type
%KWm.1.6 I	Local zone start address	
%KWm.1.7 I	Length of local zone	≥ 1 and ≤ 32 or 64 according to network type

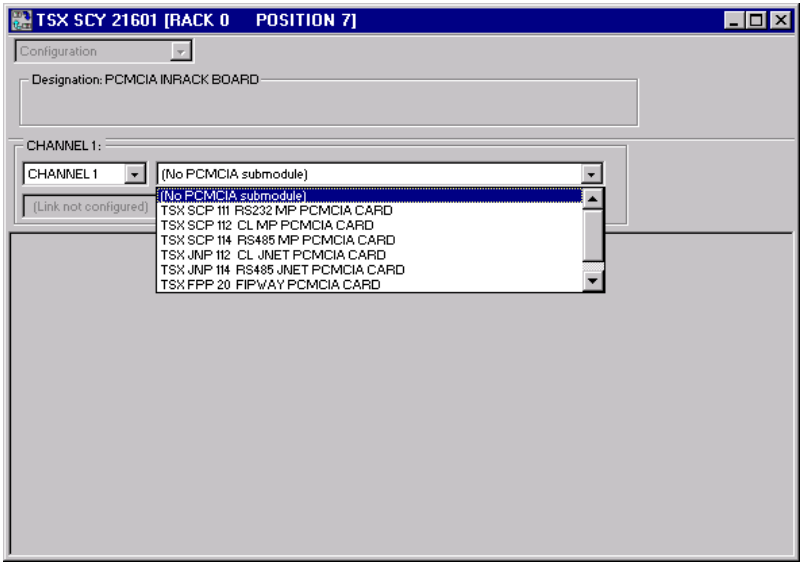
Notes : I = **I**mplicit exchanges E = **E**xplicit exchanges m = **m**odule position

3.3 Configuration Mode

3.3-1 General

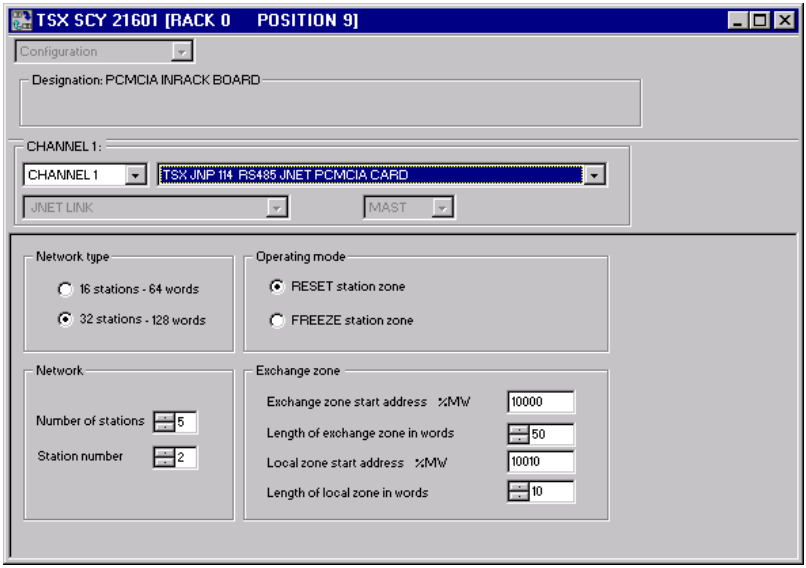
The loading and access procedures for the JNET module configuration screen are identical to those for any PCMCIA-type communication module.

Accessing the configuration of a JNET PCMCIA card :



Comment : a JNET PCMCIA card can only be inserted in channel 1 of a TSX SCY 21601 module and, as for any communication module, it is of necessity assigned in the MAST task.

3.3-2 Description of the Configuration Screen



Network type window

"16 stations - 64 words"
- Compatible with SMC 50/600 PLCs
- Number of stations ≤ 16
- Length of exchange zone ≤ 64
- Length of local zone ≤ 32

"32 stations - 128 words"
- Compatible with S1000 PLCs
- Number of stations ≤ 32
- Length of exchange zone ≤ 128
- Length of local zone ≤ 64

Note :
Times between frames are shorter in the "16 stations - 64 words" configuration than in the "32 stations - 128 words" configuration.

Network window

Number of stations

Total number of network stations
According to network type : ≤ 16 or 32

Station number

Number of the local station (being configured) according to network type : from 1 to 16 or from 1 to 32
The station number is less than or equal to the number of stations.
No two stations on the network can have the same number.

Operating mode window

Reset station zone

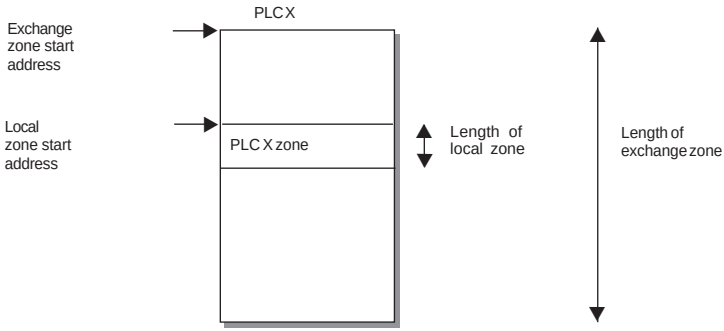
The broadcast words of a PLC are reset to zero if the PLC is absent or stopped (application program in stop mode).

Freeze station zone

The broadcast words of a PLC keep their previous state if the PLC is absent or stops (application program in stop mode).

Exchange zone window

Each PLC has a zone of 16-bit words (%MW) reserved for JNET exchanges. The zone of station X is included in the global zone. There is the same number of zones as the number of PLCs connected on the network (1 word as a minimum per station).



Comments :

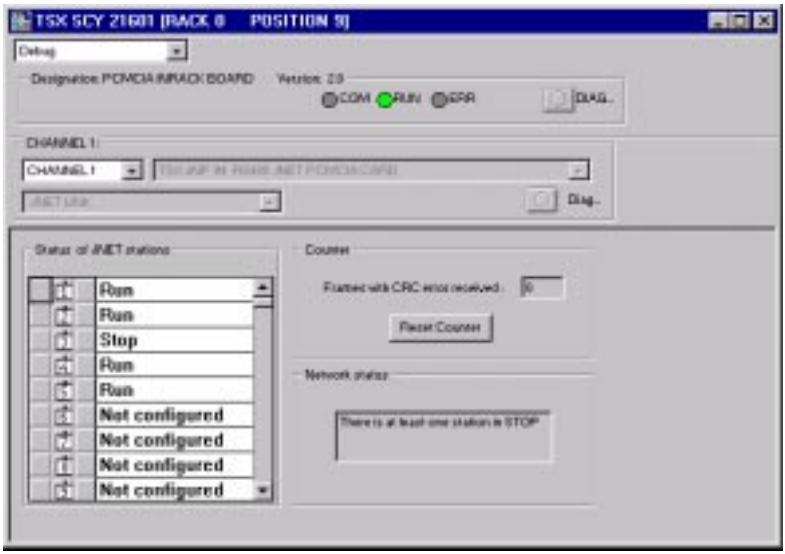
The exchange or local zone start addresses can only be entered in the form of the manufacturer's name (%MW100) : otherwise the rank of word %MW is all that is required.

3.3-3 Using JNET Exchanges

To activate JNET exchanges you must :

- Configure the various stations as shown above and, using the application program, manipulate the content of the various words produced by each station, in read and write modes.
- Transfer the various applications to the PLCs.
- Exchanges are executed without any intervention from the application program.

3.3-4 Description of the Debug Screen



Status of JNET stations

The grid on the left displays the status of the configured stations (%MWm.1.4 to %MWm.1.7).

Counter

The "Counter" zone displays the number of frames received with a CRC error (%MWm.1.3). When the counter reaches 32767, it returns to 0.
The "Reset Counter" button is used to reset this number to 0.

Network status

The "Network status" zone indicates the status (in plain text) of the special status variable (%IWm.1.0).

3.4 Operating and Fallback Modes

3.4-1 Definition of Operating Modes from a JNET Perspective

Station in Run

• The JNET and SCY modules are both OK. The application program (MAST task) is in RUN.

Station in Stop

• The JNET and SCY modules are both OK. The application program is not operating (program in Stop mode or software fault) or the application is in partial Run (one of the tasks is stopped).

Station absent

• The application program is stopped by a breakpoint or is in step by step mode.
• The CPU is faulty.
• The SCY module is faulty (no effect on JNET module).
• The JNET module is off.

To summarize :

"JNET station status"	JNET PCMCIA card	SCY	CPU	Appli. prog.
RUN	OK	OK	OK	OK (Run)
STOP	OK	OK	OK	/OK (/Run)
ABSENT	OK	OK	OK	step by step mode or breakpoint
ABSENT (*)	OK	OK	/OK	x
ABSENT (*)	OK	/OK	x	x
ABSENT	/OK	x	x	x

x : not applicable

(*) in both these cases, the JNET PCMCIA module is OK but either the CPU or the SCY is faulty (no effect on the PCMCIA card).
The JNET module is then Absent (ie. no longer participates in exchanges on the JNET network): this is identical to the behavior of the JNET S1000 module when the CPU is disconnected (or the green wire is cut).

3.4-2 Behavior of Stations in Each Case

Station in Run

• Exchanges are occurring on the network.
• The other PLCs are informed of the PLC operating mode via the network.
• The local zone, if changed, is transmitted on the network.

Station in Stop

- Exchanges are still occurring on the network.
- The station in Stop mode updates data from other stations which are in Run mode.
- The other PLCs are informed of the change in the PLC operating mode.
- The zone assigned to the PLC is processed by the other PLCs in accordance with the operating mode specified in the configuration screen (RESET or FREEZE station zones) :
  - reset data in zone to zero
  - keep the last values received
- Fallback occurs once the station is perceived to be in Stop mode.

Station absent

- If the wiring allows (especially in current loop configuration), the network continues to operate in downgraded mode.
- Fallback occurs in the same way as for a station in Stop.
- Fallback occurs once the station is perceived to be absent or transmitting invalid frames (CRC error) for three consecutive network cycles.

3.4-3 Behavior of the JNET Module when a breakpoint is set in the program

When the application program is stopped by a breakpoint or is in step by step mode, the JNET module no longer participates in exchanges on the JNET network, it is then absent for the other stations.
The ERR indicator lamp (red) of the JNET module is flashing indicating an operational fault.
To quit the step by step mode is not sufficient to restart the module exchanges. It is necessary to perform a stop command followed by a RUN command (with the programming unit or a RUN/STOP input) to remove the operational fault.

3.4-4 Behavior of the JNET Module During Backup on Memory Card

For the JNET module to restart correctly after a transfer to a Backup memory card it is necessary to perform either a PLC warm restart (power break/return) or a cold start.

Removal of the Backup memory card after a transfer automatically causes a cold start.

3.5 Performance

3.5-1 General

The JNET function ensures continuous data exchange between 32 stations on a network. This data circulates on the network in a way that is totally transparent to the user.

Data from other stations is stored in the modules as it is received and periodically sent to the CPU.

The exchange table is the same for each network station and thus constitutes a memory extension (virtual memory) for each station. It is thus important to know the maximum time required to exchange a data item.

Data is only exchanged if a change of status is apparent in the PLC broadcast zone. Detection of a change in status is managed by the JNET card and is not a function of the user program.

Once a station has detected a change in status it can only send its broadcast zone if it has the token, i.e. its right-to-send.

The graphs (section 3.5-4) show the maximum rotation time for the token as a function of the number of stations connected to the network.

The first graph shows the cycle time for the token at maximum load, i.e. when each station has detected a change of status each time it has the token. This is clearly a study case: however the graph shows the maximum time period.

The second graph is more representative of reality. In this case it is assumed that on average each station detects a change of status every second.

Comment :

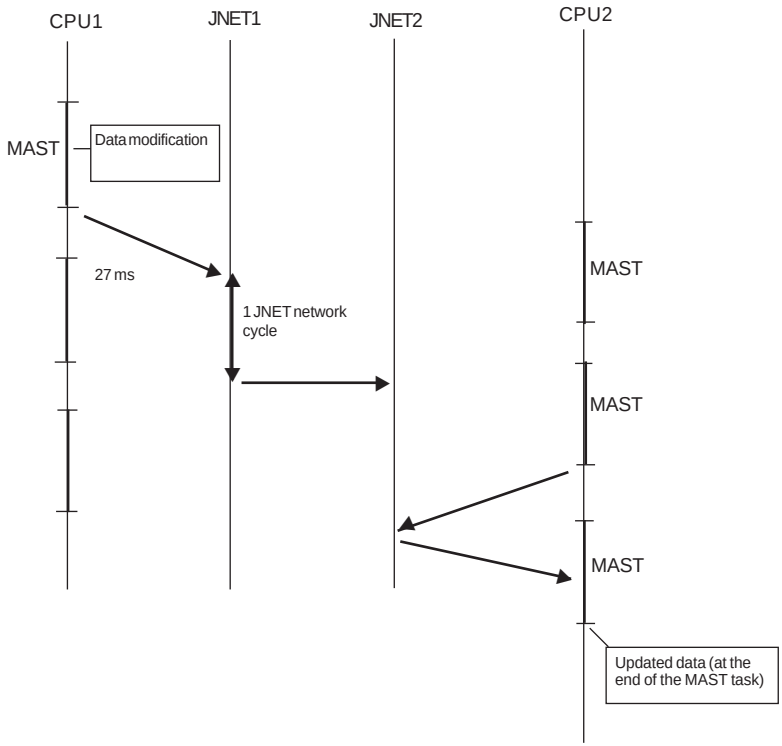
The graphs (section 3.5-4) only show data transmission time on the JNET network itself. The time to update the data in the PLC memory must be added to this (section 3.5-3).

3.5-2 Impact of a JNET Channel on the PLC Scan Time

The impact of a JNET function on the MAST task scan time is approximately 5 ms maximum.

3.5-3 Response Time Between Applications

Exchanges between the CPU and the JNET module are via messages.
The response time between applications is as follows (worst case scenario) :



The maximum response time between applications is therefore :

27 ms + JNET network cycle time + 2 MAST scans (on condition that the MAST task is longer than 30 ms).

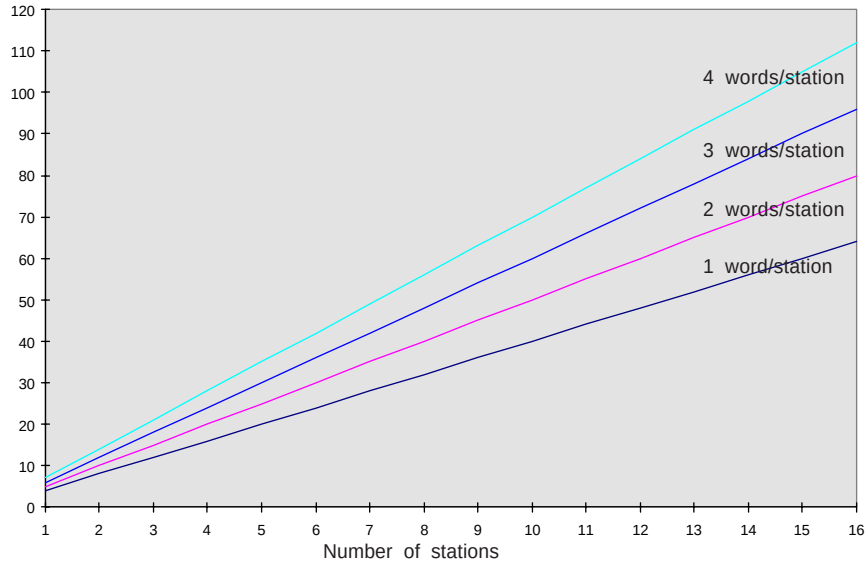
Note :
The JNET network cycle time represents the rotation time for the token (sections 3.5-1 and 3.5-4).

3.5-4 Graphs

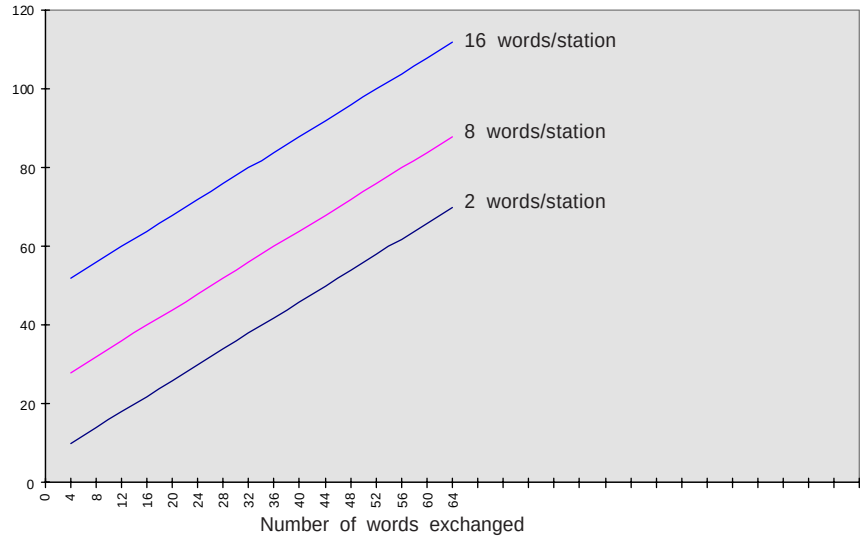
• Network type "16 stations / 64 words"

Maximum load : Words changed on each rotation of the token

Token cycle time (ms)



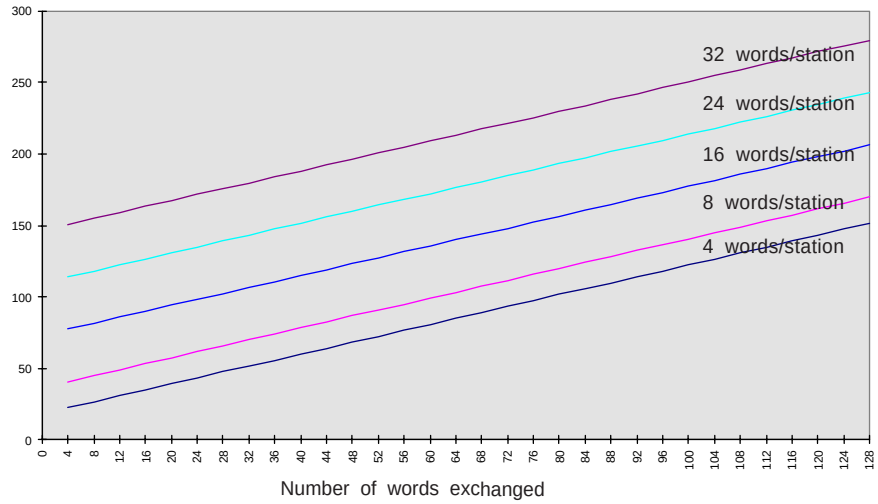
Average token cycle time (ms)



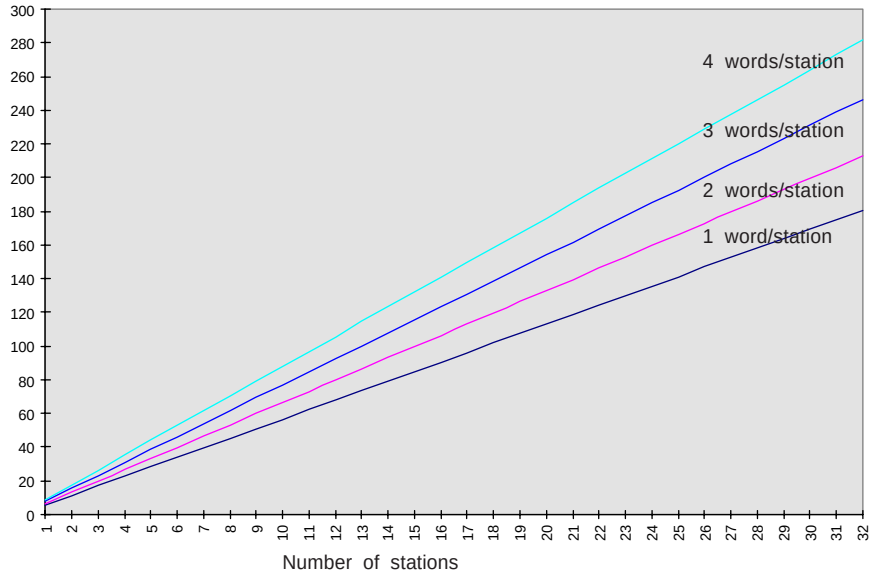
• Network type "32 stations / 128 words"

Maximum load : Words changed on each rotation of the token

Average token cycle time (ms)



Token cycle time (ms)



• 4 words exchanged per station every second

